

Snehal Gore

Los Angeles, CA | ssgore18@gmail.com | +1 310-462-7354 | linkedin.com/in/snehal-gore | github.com/snehalgore1 | snehalgore1.github.io

SUMMARY

Software Engineer with 2.5+ years shipping production systems; M.S. in Computer Science (AI) from USC, December 2026; available full-time January 2027 and open to relocation. Backend services, distributed systems, AI/ML infrastructure, and LLM applications.

EXPERIENCE

SnapRefund (Early-Stage InsurTech Startup)

Los Angeles, CA

Software Engineer Intern

May 2026 – Aug 2026

- Built **AiSnap**, a production **Gemini function-calling** assistant that lets merchants and admins query live payment data in natural language, translating questions into typed **PostgreSQL** tools via a **NestJS** API.
- Reduced ungrounded financial answers from **45% to 5%** by detecting responses that reported figures without a database call and forcing tool-use regeneration; added tenant-scoped queries, field allowlisting, PII sanitization, and burst/daily rate limiting.
- Led migration of a payments platform from Firebase Cloud Functions to NestJS/PostgreSQL, re-implementing Firebase's script password hashing so **10K+ existing users** authenticated with **zero password resets**.
- Engineered concurrency-safe payment processing with PostgreSQL **SKIP LOCKED** row locking and stale-lock recovery to horizontally scale mass-payment workers, using idempotency keys to guarantee **exactly-once** recurring disbursements.
- Built signed outbound webhook delivery with HMAC-SHA256 signatures, exponential retry, and per-account event subscriptions on **AWS SQS**, routing failures through a dead-letter queue to SNS/Lambda alerts, provisioned in Pulumi across 4 environments.

Whirlpool Corporation

Pune, India

Software Engineer, Machine Learning

Jul 2022 – Dec 2024

- Migrated production ML inference from cloud to on-device execution across **1M+ connected appliances** with quantized TensorFlow Lite models, eliminating **\$250K+/year** in cloud costs and cutting latency **3×**.
- Replaced rule-based dryer time-remaining logic with a **sub-1K-parameter** neural network after LSTM, XGBoost, and Random Forest exceeded controller memory, deployed via TinyML to 1M+ dryers, cutting error **7×** (± 35 to ± 5 min).
- Built the oven's lens-cleanliness detection in the Android touchscreen app (Java, CameraX, TensorFlow Lite) with INT8-quantized EfficientNet on a 4×4 frame grid, prompting a clean at 75%+ obstruction; won a **Whirlpool STAR Award**.
- Automated build, packaging, and staged **OTA rollout** of on-device models across cooking and laundry product lines, cutting release cycles from **days to hours** while sustaining **99.5%** latency-SLA compliance.
- Generated **50K+** synthetic stain images by fine-tuning Stable Diffusion with DreamBooth, raising a 20+ class classifier's accuracy **38%**; fine-tuned **Mistral-7B with QLoRA** on 8K+ fabric-care prompt pairs for wash recommendations.

PROJECTS

Distributed Object Store | C++20, gRPC, Raft, Kubernetes

github.com/snehalgore1/distributed-object-storage

- Built a C++20 distributed object store with a consistent-hash ring, RF=3 replication over gRPC, and W=2 quorum writes, sustaining about **3.8K req/s** on a 3-node cluster; implemented **Raft from scratch** for the metadata control plane.
- Made writes atomic and fsync'd behind a CRC32-checked write-ahead log with SHA-256 verified reads; validated by **131 tests** including chaos tests that kill nodes and corrupt data, with ThreadSanitizer CI.

MiniTensorRT: C++ Inference Runtime | C++17, CUDA, ARM NEON

github.com/snehalgore1/mini-tensorrt

- Built a from-scratch inference runtime with a graph IR, fusion passes, and CPU and CUDA backends, running GPT-2 (124M) token-for-token identical to Hugging Face; a 64-token forward pass takes **21.7 ms** on a Tesla T4.
- Raised GEMM throughput about **180×** (1.7 to 310 GFLOP/s) with a packed ARM NEON microkernel, cutting GPT-2 block latency **56×** and peak memory **12.5×** through arena memory planning and operator fusion.

Agentic RAG and Hybrid Search Platform | Python, LangGraph, OpenSearch, Airflow

github.com/snehalgore1/agentic-rag

- Improved Recall@10 by **35%** over BM25-only on 10K+ documents by combining BM25 and dense retrieval with Reciprocal Rank Fusion, at **under 2 s p95** across 7 containerized services, with a LangGraph agent for relevance grading and query rewriting.

EDUCATION

University of Southern California

Los Angeles, CA

M.S. Computer Science, AI Specialization

Jan 2025 – Dec 2026 (Expected)

Coursework: Analysis of Algorithms, Machine Learning, Applied NLP, Information Retrieval, Hardware Foundations of ML

Savitribai Phule Pune University

Pune, India

B.E. Computer Engineering; GPA 9.4/10

May 2018 – May 2022

TECHNICAL SKILLS

Languages: Python, C++ (17/20), TypeScript, JavaScript, SQL, Java, CUDA C++, Bash

Backend & Systems: NestJS, Node.js, FastAPI, gRPC, PostgreSQL, Redis, OpenSearch, Raft, concurrency, distributed systems

AI/ML: PyTorch, TensorFlow Lite, Hugging Face, ONNX, LangGraph, RAG, LoRA/QLoRA, function calling, quantization

Cloud & DevOps: AWS (SageMaker, S3, SQS, Lambda, Redshift), GCP Vertex AI, Docker, Kubernetes, Pulumi, MLflow, Airflow

Awards: Whirlpool STAR Award (2023); 1st place of 200+ teams, Whirlpool Global Software Hackathon (2024)